

Addendum: Automated Scientific Discovery Integration

Extending EVOCOMP with AI-Scientist for Meta-Scientific Loop Closure

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*Addendum to: EVOCOMP: An Evolutionary Compiler Architecture
for Scalable, Verifiable, and Economical Neural Model Composition*

Abstract

This addendum extends the EVOCOMP dissertation (January 2026) to include integration with **AI-Scientist**, an open-source automated scientific discovery system developed by SakanaAI. While the core EVOCOMP architecture operates independently, AI-Scientist provides a meta-cognitive layer for hypothesis generation, experimental design, and automated paper writing that synergizes with EVOCOMP’s verification and compilation pipeline. We detail technical integration points, updated economic models, and theoretical implications for recursive self-improvement in AI research systems. The integration increases total costs by 12.9% (\$477) while eliminating manual hypothesis engineering and enabling continuous automated research cycles.

1 Introduction

Following the completion of the primary EVOCOMP dissertation [EVOCOMP, 2026], we identified **AI-Scientist** [Lu et al., 2024]—an open-source automated scientific discovery system developed by SakanaAI—as a complementary framework for enhancing the EVOCOMP feedback loop.

AI-Scientist employs Large Language Models (LLMs) as autonomous research agents capable of:

1. **Idea Generation:** Proposing novel research directions based on literature review and code analysis
2. **Experimental Design:** Automatically generating and executing experimental protocols
3. **Paper Writing:** Producing publication-ready manuscripts with LaTeX formatting

This addendum specifies the integration architecture, economic implications, and safety considerations for combining EVOCOMP’s evolutionary compiler with AI-Scientist’s automated discovery capabilities.

2 Technical Overview

AI-Scientist operates through a multi-agent pipeline: *Generation* $\rightarrow$ *Review* $\rightarrow$ *Revision*, mirroring the peer-review process but automated at machine speed. The system utilizes Claude 3.5 Sonnet or GPT-4o as base models, with costs averaging \$2–4 per paper generation.

3 Integration with EVOCOMP Architecture

We propose AI-Scientist as a **Phase 4** component of the EVOCOMP pipeline—the *Meta-Scientific Loop*—extending the original three-phase architecture:

Phase	EVOCOMP Component	AI-Scientist Role
1	Distributed Discovery	Hypothesis generation for manifold geometries
2	Verification	Automated theorem formulation for Z3 specs
3	Serving	Anomaly interpretation and trigger analysis
4	Meta-Learning	Automated experimental design

Table 1: Integration points between EVOCOMP and AI-Scientist

3.1 Automated Hypothesis Generation

AI-Scientist can operate on EVOCOMP’s archive of evolved models to generate hypotheses about unexplored regions of the manifold:

```

1 class AutomatedEvoComp:
2     """
3     EVOCOMP enhanced with AI-Scientist meta-controller.
4     """
5     def __init__(self):
6         self.evocomp = EVOCOMPRuntime()
7         self.ai_scientist = AIScientistAgent(
8             template_dir="templates/evo_comp",
9             model="claude-3-opus-20240229"
10        )
11
12    def meta_scientific_loop(self, iterations: int = 100):
13        """
14        Automated closed-loop research.
15        """
16        for i in range(iterations):
17            # AI-Scientist analyzes current archive
18            hypothesis = self.ai_scientist.generate_idea(
19                context={
20                    "current_pareto_front": self.evocomp.archive.
21                        fitnesses,
22                    "manifold_coverage": self.evocomp.manifold.
23                        coverage(),
24                    "recent_failures": self.evocomp.failure_log
25                }
26            )

```

```

25
26     # EVOCOMP validates experimentally
27     experiment = self._design_experiment(hypothesis)
28     results = self.evocomp.execute(experiment)
29
30     # Automated write-up if significant
31     if results.significance_p < 0.05:
32         paper = self.ai_scientist.write_paper(
33             idea=hypothesis,
34             results=results,
35             template="neurips_2024"
36         )
37     self._submit_to_arxiv(paper)

```

3.2 Safety Specification Synthesis

A critical integration point involves automated generation of safety specifications for the Z3 verification layer. AI-Scientist can analyze failure modes from the Ethical Circuit Breaker to propose formal safety constraints:

Theorem 3.1 (Automated Safety Specification). Given a log of guard violations $\mathcal{V} = \{(x_i, y_i, \text{violation}_i)\}_{i=1}^n$, AI-Scientist can generate candidate safety specifications $\mathcal{S}_{\text{cand}}$ such that:

$$\forall s \in \mathcal{S}_{\text{cand}}, \quad \Pr[\text{violation}|s] < \epsilon \quad (1)$$

with verification via EVOCOMP’s Z3 backend.

Proof. AI-Scientist performs inductive logic programming on the violation log to hypothesize constraints $\mathcal{S}_{\text{cand}}$ that separate violating from non-violating regions of the manifold. EVOCOMP’s Z3 integration then verifies these constraints formally, rejecting candidates where $\exists x : \text{violation}(x) \wedge s(x)$ is satisfiable. $\square$

4 Economic Analysis

The integration affects the economic analysis presented in the main dissertation:

Component	Original Cost	With AI-Scientist
Discovery	\$3,024	\$3,024 (unchanged)
AI-Scientist API	\$0	\$500 (LLM tokens)
Automated Verification	\$100	\$50 (reduced human oversight)
Paper Writing	Manual	Automated (sunk cost reduction)
Total	\$3,697	\$4,174

Table 2: Updated cost model including AI-Scientist integration

The modest increase (\$477, +12.9%) is offset by:

- Elimination of manual hypothesis engineering (40 hours $\rightarrow$ 0 hours)
- Automated literature review and citation management
- Continuous experimental monitoring without human intervention

5 Theoretical Extensions

5.1 Recursive Self-Improvement

The combination of EVOCOMP and AI-Scientist enables a limited form of recursive self-improvement:

Definition 5.1 (Constrained Recursion). Let $\mathcal{E}_t$ be the EVOCOMP system at generation t , and $\mathcal{A}$ be the AI-Scientist agent. The system evolves as:

$$\mathcal{E}_{t+1} = \mathcal{E}_t \oplus \mathcal{A}(\text{analysis}(\mathcal{E}_t)) \quad (2)$$

where $\oplus$ denotes safe composition via manifold merging, constrained by the formal verification layer to prevent value drift.

5.2 The Automated Scientist as a Niche

Within EVOCOMP’s ensemble architecture, AI-Scientist can be viewed as a specialized *meta-niche*—a model responsible for high-level research strategy rather than task execution:

```

1 class MetaScientistNiche(EvoManifoldKernel):
2     """
3     AI-Scientist as an evolved niche in the ensemble.
4     """
5     def forward(self, research_state: ResearchState) ->
6         ResearchAction:
7         """
8         Determines next experimental steps.
9         """
10        # Geodesic distance to successful prior experiments
11        similarity = self.manifold.distance(
12            research_state.embedding,
13            self.successful_experiments
14        )
15        if similarity > self.exploration_threshold:
16            return ResearchAction.EXPLORE_NOVEL
17        else:
18            return ResearchAction.EXPLOIT_CURRENT

```

6 Safety and Ethical Considerations

The integration introduces new safety considerations beyond those addressed in the main dissertation:

1. **Hypothesis Validation:** AI-Scientist generated hypotheses must pass through EVOCOMP’s formal verification layer before experimental execution to prevent unsafe experimental designs.
2. **Publication Safety:** Automated paper writing requires automated fact-checking to prevent the dissemination of false or misleading results.

3. **Recursive Limits:** While constrained recursion is enabled, hard limits on modification of the safety verification layer itself prevent bootstrap attacks.

7 Implementation Notes

The integration requires minimal changes to the existing EVOCOMP codebase:

- **Dependency:** `pip install ai-scientist` (SakanaAI package)
- **API Key:** Requires `ANTHROPIC_API_KEY` or `OPENAI_API_KEY` environment variable
- **Templates:** Custom LaTeX templates in `templates/evo_comp/` directory
- **Hook:** Add `MetaScientificLoop` class to `evocomp/serving/` module

8 Conclusion

While the core EVOCOMP contribution stands independently, AI-Scientist provides a natural extension for automating the scientific validation loop. This integration aligns with the dissertation’s vision of democratizing AI research by reducing both computational costs (via EVOCOMP) and human labor costs (via AI-Scientist).

Future work should explore the safety implications of closed-loop automated discovery, particularly regarding the verification of AI-generated hypotheses before experimental execution.

References

- Lu, C., Lu, C., Lange, R. T., Foerster, J., Clune, J., & Ha, D. (2024). The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv preprint arXiv:2408.06292*.
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